

سنوات: صيفي 2020

1)) The value of the Integral of $\int x \sinh x \, dx$ is equal to :-

$$\begin{aligned} F(x) &= x & G'(x) &= \sinh x \, dx \\ F'(x) &= 1 & G(x) &= \cosh x \, dx \end{aligned}$$

$$x \cosh x - \int \cosh x \, dx \rightarrow \sinh x$$

$$(x \cosh x - \sinh x) \Big|_0^3$$

$$(3 \cosh 3 - \sinh 3) - (0 \cosh 0 - \sinh 0)$$

$$\Rightarrow (3 \cosh 3 - \sinh 3)$$

zero

2)) Evaluate the Integral $\int (\cosh^2 x - \sinh^2 x) \, dx$?

$$\int_0^3 (\cosh^2 x - \sinh^2 x) \, dx \quad \text{متطابقة = 1}$$

$$\int_0^3 1 \, dx \rightarrow x \Big|_0^3 \rightarrow 3 - 0 = \underline{\underline{3}}$$



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3)) $f(x) = (x+1) \sinh(3x)$ then $f'(0) =$

$$f(x) = (x+1) \sinh(3x)$$

$$f'(x) = ((x+1) \cosh(3x) * 3) + (\sinh(3x) * 1)$$

$$(0+1) \cosh(0) * 3 + \sinh(0) * 1$$

$$1 * 3 = \underline{3}$$

4)) If $y = \underline{3}^x + \ln x$... then $\frac{dy}{dx} =$

$$y' = \underline{3}^x \ln 3 * 1 + \frac{1}{x} \rightarrow 3^x \ln 3 + \frac{1}{x}$$

له القول نفسه * لو تبت * مشتقة الدالة

5)) is the following polynomials prime or not ?

1 $x^2 + x + 1 \rightarrow \text{prime} \checkmark$

2 $x^2 + 3 \rightarrow \text{prime} \checkmark$

* أي اقتراف فوقه تربيعي غالباً

3 $x + 1 \rightarrow \text{prime} \checkmark$

* غير أولي (not prime)

4 $x^4 + 1 \rightarrow \text{prime} \checkmark$



*6)) which of the following is improper integral :-

1 $\int_0^{\frac{\pi}{6}} \csc x \, dx$ improper $\rightarrow \left(\csc \rightarrow \frac{1}{\sin} \rightarrow \frac{1}{\sin 0} = \frac{1}{0} \right)$

2 $\int_2^3 \ln x \, dx$ not improper $\left. \begin{array}{l} \ln 2 = \ln 0 \\ \ln 3 = \ln 0 \end{array} \right\} \rightarrow \text{define } \checkmark$

7)) Evaluate the Integral $\int \frac{dx}{4+x^2}$?

$x = 2y \rightarrow dx = 2dy$

$\Rightarrow \int \frac{2dy}{4+4y^2} \rightarrow \int \frac{2dy}{4(1+y^2)} = \frac{1}{2} \int \frac{1}{1+y^2} dy$

$\tan^{-1} y$

$\frac{1}{2} \tan^{-1} y \rightarrow \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C$

8)) If $y = \sin^{-1}(4x)$ then $dy/dx = ?$

$\frac{4}{\sqrt{1-16x^2}}$

الطريقة

$\frac{a}{\sqrt{1-(ax)^2}}$

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التأخذ

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9)) Simplify the expression $e^{-\frac{1}{2} \ln x}$

$$\Rightarrow e^{\ln x^{-\frac{1}{2}}} \rightarrow e^{\ln x^{-\frac{1}{2}}} = x^{-\frac{1}{2}} = \frac{1}{x^{\frac{1}{2}}} = \frac{1}{\sqrt{x}}$$

10)) Compute the integral $\int 6 \sec^2 x \tan^5 x \, dx$?

$$\tan x = y \rightarrow \sec^2 x = \frac{dy}{dx}$$

$$\int 6 \sec^2 x y^5 \frac{dy}{\sec^2 x} = \int y^5 dy$$

$$6y^6 + C \rightarrow 6 \tan^6 x + C$$

11)) $\lim_{x \rightarrow 0} \frac{10e^x - 10x - 10}{x^2} = ?$ (لوبيتا إلى 0/0)

$$\lim_{x \rightarrow 0} \frac{10e^x - 10}{2x} = \frac{0}{0} \quad \text{لوبيتا إلى 0/0}$$

$$\lim_{x \rightarrow 0} \frac{10e^x}{2} = 5$$



12)) Using Trigonometric substitution to evaluate the integral $\int \frac{\sqrt{x^2-9}}{x^2} dx$, then $x = ?$

$$x = 3 \sec \theta \rightarrow \sqrt{x^2-9}$$

$\searrow$
 $\sqrt{9}$

13)) If the series $\sum_{k=1}^{\infty} (5+a_k)$ converge to 8 , then $\lim_{k \rightarrow \infty} a_k = ?$

equal to 8 (sum of $\sum_{k=1}^{\infty} (5+a_k)$)

14)) If S_4 is the 4th partial sum of the series $\sum_{k=1}^{\infty} (-1)^{k+1}$ then $S_4 = ?$

First partial when $k=1 \rightarrow$ sec $(k=2) \rightarrow$ third $(k=3) \rightarrow$ fourth $(k=4)$

$$\sum_{k=1}^{\infty} (-1)^{k+1} \rightarrow (-1)^{4+1} \rightarrow -1^5 = -1$$

$$S_4 = -1$$



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15)) If the interval of $\sum_{k=1}^{\infty} a_k (x-a)^k$ is $[-2, 6)$, then the radius of the convergens is ...

$$\frac{6 - (-2)}{2} = \frac{6+2}{2} = \frac{8}{2} = 4$$

16)) If $(a < 0)$, then $\lim_{x \rightarrow \infty} (8x e^{ax})$

$$a < 0 \rightarrow \left(\frac{8x}{e^{-ax}} = \frac{1}{e^{ax}} \right) \quad \text{ننزل السالب المقام}$$

$$\frac{8x}{e^{ax}} \rightarrow \frac{8}{a e^{ax}} = \frac{8}{\infty} = \text{Zero}$$

17)) The range of $f(x) = 6 \tan^{-1}(x)$ is :-

a $(-3\pi, 3\pi)$ b $[-3\pi, 3\pi]$ c $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

18)) $\lim_{x \rightarrow \infty} \frac{\ln(x+1)}{\ln(3x+1)} = \frac{\infty}{\infty} !$ (لوبيتال)

$$\Rightarrow \frac{\frac{1}{x+1}}{\frac{3}{3x+1}} = \frac{\infty}{\infty} ! \rightarrow \frac{3x+1}{3(x+1)} = \frac{3}{3} = 1$$



19)) Let $\{a_n\}$ be the sequence defined by : $a_1 = 0$, $a_2 = 2$ and $a_n = a_{n-1} + a_{n-2}$, then a_5 equal ?

✓
a. 6 b. 4 c. 16 d. 11

20)) Which of the following polynomials is not prime ?

a $x^2 + x + 16$ prime ✓

b $x + 16$ prime ✓

c $x^4 + 16$ not prime x

d $x^2 + 16$ prime ✓

21) IF S_4 is the 4th partial sum of the series $\sum_{k=1}^{\infty} (-1)^k$ then $S_4 =$?

$$S_1 (k=1) = (-1)^1 = -1$$

$$S_2 (k=2) = (-1)^2 = 1$$

$$S_3 (k=3) = (-1)^3 = -1$$

$$S_4 (k=4) = (-1)^4 = 1$$



22)) Evaluate the Integral $\int_1^5 (\cosh^2 x - \sinh^2 x) dx$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\int_1^5 1 dx \rightarrow x \Big|_1^5 = 5 - 1 = 4$$

23)) The suitable assumption for $\int \frac{\sqrt{25-x^2}}{x^2} dx$ is ...

* حرق بقة داخل الجذر $\sqrt{1-x^2}$ نفرضنا عو تسيدج ازا 2 عاقل

مسترك به + السك الى العا ٢ جب *

$$* 5 \sin \theta$$

24)) $\int \sinh\left(\frac{x}{2}\right) dx \rightarrow ?$

$$\frac{1}{2} \cosh\left(\frac{x}{2}\right) + C$$

25)) If $y = \tan^{-1}(x)$, then $\frac{dy}{dx} = ?$

$$\rightarrow \left(\frac{1}{1+x^2} \text{ مكنوت} \right)$$

26)) If $y = \cot^{-1}(x)$, Then $\frac{dy}{dx} = ?$

$$\rightarrow \left(\frac{-1}{1+x^2} \text{ مكنوت} \right)$$



$$* \text{Test } \sum_{n=1}^{\infty} \frac{\sqrt{k+1}}{\sqrt{k^3+2}} \quad (\text{limit conv. Test}) \Rightarrow \underline{\text{div}}$$

$$* \text{Test } \sum \frac{9}{3^n + 5} \quad (\text{limit conv. Test}) \Rightarrow \underline{\text{conv}}$$

$$* \text{Test } \sum_{n=1}^{\infty} \frac{1}{n+3^n} \quad (\text{comparison test}) \Rightarrow \underline{\text{conv}}$$

$$* \text{Test } \sum \frac{(2n+1)^n}{n^{2n}} \quad (\text{Root test}) \Rightarrow \underline{\text{conv}}$$

$$* \text{Test } \sum \left(\frac{3n^2+1}{2n^2+1} \right)^n \quad (\text{Root test}) \Rightarrow \underline{\text{div}}$$

$$* \text{Test } \sum \frac{2n+3}{2n^2+4} \quad (\text{Root test}) \Rightarrow \underline{\text{div}}$$

$$* \text{Test } \sum \frac{2^k k!}{(k+2)!} \quad (\text{Ratio test}) \Rightarrow \underline{\text{div}}$$

$$* \text{Test } \sum_{n=1}^{\infty} \frac{e^{\frac{1}{n}}}{n} \quad (\text{limit comparison test}) \Rightarrow \underline{\text{div}}$$

$$* \text{test } \sum \frac{(-1)^n}{n^2} \quad (\text{Alternative series}) \Rightarrow \underline{\text{conv}}$$

$$* \text{test } \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3+4} \quad (\text{Alternative Series}) \Rightarrow \underline{\text{conv}}$$



* $\lim_{x \rightarrow \infty} (8x e^{ax}) = \underline{\hspace{2cm}}$ If $a < 0$? (Zero)

* If S_5 is the 5th partial sum of the series $\sum_{k=1}^{\infty} (-1)^{k+1}$
Then $S_5 = \underline{\hspace{2cm}}$? (1)

* The value of the Integral $\int_0^3 x \sinh x \, dx$ is equal to...
(3 cosh - sinh 3)

* $\{5 + \sqrt{n}\}_{n=1}^{\infty}$ The sequence is...
a diverges b converges to 6 ✓

b converges to 7 d converges to 5

* $\int_0^4 \cosh^2 x - \sinh^2 x \, dx$ is equal to ?
answer = 3

* Find $\sum_{n=0}^{\infty} \frac{10^n}{(-9)^{n+1}}$?

$\sum \left(\frac{-1}{9}\right) \left(\frac{10}{-9}\right)^n \rightarrow r = \left(\frac{10}{9} > 1\right)$ div.



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* Find $\sum_{n=5}^{\infty} 2 \left(\frac{1}{4}\right)^n$ (Geometric)

* test $\sum \tan^{-1}\left(\frac{3^n}{3^{n+2}}\right)$ (limit test!) $\Rightarrow$ Div

* $\int_{-\infty}^{\infty} \frac{2}{4+x^2} dx$? $3\pi/4$

* $\sum_{n=1}^{\infty} 2^{1/n}$? answer = $e^0 = 1 = \text{div}$

* $\int_{-1}^2 \frac{1}{\sqrt{4-x^2}} dx$? $(\cos^{-1} \frac{x}{2}) \dots$ answer $-\frac{\pi}{3}$

* the domain of $\sec^{-1}$ is ... ? $|x| > 1$

* If $f(x) = x \ln x$ then $f''(x) = ?$

$f' = x \left(\frac{1}{x}\right) + \ln x = 1 + \ln x$

$f'' = \frac{1}{x}$

* $\sum_{n=1}^{\infty} \frac{\cos n\pi}{n}$ $\Rightarrow$ Converge

